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CASE STUDY

IOT SCADA SYSTEM FOR PM KUSUM

INDUSTRY	Renewable Energy
OBJECTIVE	PM KUSUM Yojana to increase the income of farmers, provide sources of irrigation & cut down fossil fuel consumption
PROJECT	IoT SCADA system for PM Kusum

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CHALLENGES

- **SEDM Portal Integration:** Update MNRE Data with regard to installed capacity
- **Seasonal Variations:** Sunlight availability & intensity not consistent across seasons and geographies affecting energy output
- **Long-Term Sustainability:** Ensuring the long-term operation, maintenance, and efficiency of installed solar systems
- **Dust and Maintenance Issues:** Dust reduces efficiency of solar panels requiring regular cleaning and maintenance
- **Distributed Architecture:** Managing multiple solar plants under a single portal is challenging. Hence, need for centralized integration





- **Security and Compliance:** Cybersecurity compliance & secure data transmission to government organizations
- **Active and Reactive power Control:** Efficient control essential for optimizing power generation and to ensure grid stability
- **Lack of Awareness:** Farmers are not fully informed about the benefits of PM-KUSUM viz. application process leading to lower adoption rates
- **Connectivity Issues:** Cable infrastructure unavailability leads to data reliability concerns as communication depends solely on internet connectivity

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SOLUTION



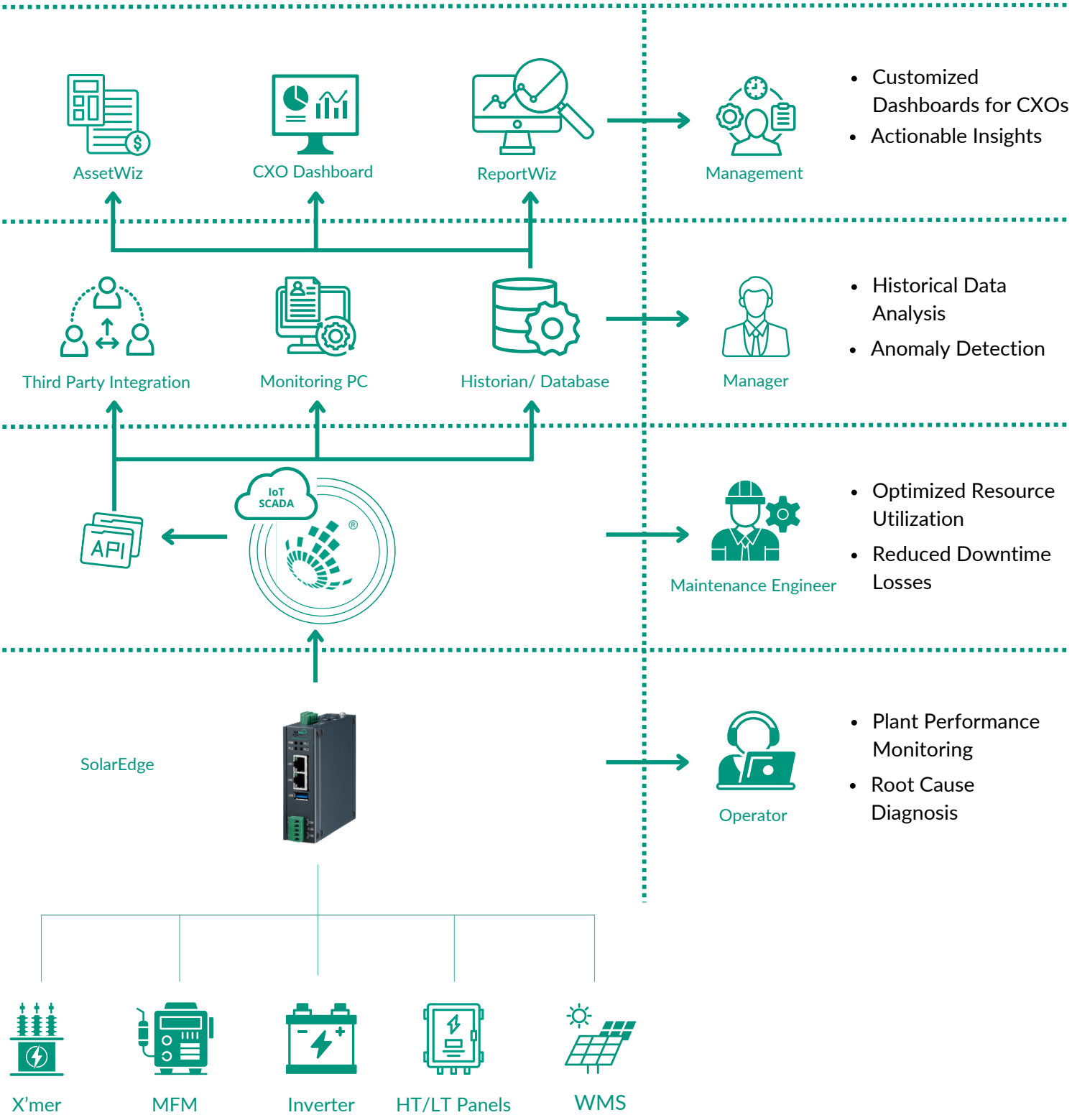
- **Temperature Degradation:** Generation loss due to temperature of the equipment gets. checked
- **Security and Compliance Issues:** Cyber Security Compliant System architecture ensures compliance
- **IOT Controller:** An IOT based Edge Computing Device for acquiring data from Renewable Assets
- **CXO Dashboard:** Plant KPI Parameters with Targeted vs Actual KPI Parameters
- **SEDM Portal Integration:** SEDM Portal Integration as per required data in required format with timestamp
- **Multi-Plant Configuration and O&M:** Supports multiple and geographical assets through a common control centre

- **Energy and Loss Calculation:** Calculates the generation loss at the inverter end and billing MFM
- **Predictive and Prescriptive Maintenance Using AI-ML Technology:** AI-based loss prediction, and automated loss alerts
- **Zero Data loss:** In the event of unavailability of network, IOT Controller stores the data and forward to cloud system in the event of network unavailability
- **String Level Monitoring:** String level monitoring with cleaning cycle integration made available on the portal to avoid generation loss
- **Solar Pro and Solar Edge:** Comprises Intelligent Data Pre-Processing and Synchronizing it with Solar Pro while SolarEdge stores and forwards to ensure zero data loss



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ARCHITECTURE



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VALUE CREATION

- **Power Optimization:** Automated Active Power and Reactive Power control
- **Security & Compliance:** End-to-end encryption, regulatory compliance and periodic security audits
- **Energy Loss Reduction:** Real-time energy auditing, AI-based loss prediction, and automated loss alerts
- **SEDM & ADMS Integration:** Automated data formatting, timestamp synchronization, & standardized structures
- **Sustainability & Efficiency:** IoT-based predictive maintenance, automated O&M scheduling & remote monitoring





- **Dust & Maintenance:** String-level monitoring, AI-powered cleaning alerts, and self-cleaning panels
- **Increased Farmers' Income:** Lower electricity costs increase farmers' overall income
- **Environmental Protection:** Farmers contribute to environmental protection by utilizing green energy
- **Multi-Plant Management:** Centralized dashboard, scalable cloud infrastructure, and role-based access control
- **Connectivity Solutions:** IoT controllers with offline storage, edge computing, and multiple communication protocols

21st Floor, "Yash Anant",
Opp. Old RBI, Ashram Road,
Ahmedabad - 380009,
Gujarat, India

+917969269000

info@adaptive-engg.com